IRRG1816C Series Datasheet



Features & Benefits

- Package: SMT package 1816（1.85*1.65*0.6mm）
- Chip Material: IR: AlGaAs/Si, R: AlGaInP/Si, G: GaN/Si
- Wavelength: IR: 940nm, R: 660nm, G: 525nm
- Optical Power Class: IR: 3.0mW/sr@20mA, R: 300mcd@20mA, G: 2200mcd@20mA
- Luminous surface appearance: Water Clear
- Soldering methods: IR Reflow

Typical Application:

- Infrared applied system

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Table of Contents

1. Absolute Maximum Ratings(Ta=25°C) 3
2. Electrical-optical characteristics(Ta=25°C) 4
3. Product Binning 5
4. Package Outline Dimension 6
5. Tapping specifications (unit: mm) 7
6. Package Method:(unit: mm) 7
7. Label description 8
8. Typical Electro-Optical Characteristics Curves 9
9. Reliability test items and conditions 12
10. Precautions for use 13

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1. Absolute Maximum Ratings(Ta=25°C)

Parameter	Symbol	Rating	Unit
Soldering Temperature	Tsol	Reflow soldering (260°C for 10 sec.) Hand soldering(300°C for 3sec)	°C
Operating Temperature	Topr	-30°C ~ +85°C	-
Storage Temperature	Tstg	-40°C ~ +100°C	-
Infrared Emitter (IR)			
Power Dissipation	P_C	105	mW
Forward Current*	I_F	60	mA
Peak Forward Current*	I_{FP}	300	mA
Reverse Voltage	V_R	5	V
Red Emitter (R)			
Power Dissipation	P_C	95	mW
Forward Current*	I_F	40	mA
Peak Forward Current*	I_{FP}	200	mA
Reverse Voltage	V_R	5	V
Green Emitter (G)			
Power Dissipation	P_C	80	mW
Forward Current*	I_F	30	mA
Peak Forward Current*	I_{FP}	200	mA
Reverse Voltage	V_R	5	V

❶ IF condition: for 10 seconds or less.

❷ IFP condition: pulse width ≤1msec, duty cycle ≤1/10.

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2. Electrical-optical characteristics(Ta=25°C)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Infrared Emitter (IR)						
Radiant Intensity	I_E	1.5	3.0	-	mW/sr	I_F=20mA
Forward Voltage	V_F	-	1.3	1.5	V	
Peak Wavelength	λ_P	-	940	-	nm	
Spectral Bandwidth	$\Delta\lambda$	-	30	-	nm	
Reverse Current	I_R	-	-	10	μA	V_R=5V
View Angle	$2\theta_{1/2}$	-	120	-	deg	I_F=20mA
Red Emitter (R)						
Luminous Intensity	I_V	260	300	-	mcd	I_F=20mA
Radiant Intensity	I_E	2.5	4		mW/sr	
Forward Voltage	V_F	-	2.0	2.3	V	
Peak Wavelength	λ_P	-	660	-	nm	
Spectral Bandwidth	$\Delta\lambda$	-	30	-	nm	
Reverse Current	I_R	-	-	10	μA	V_R=5V
View Angle	$2\theta_{1/2}$	-	120	-	deg	I_F=20mA
Green Emitter (G)						
Luminous Intensity	I_V	1400	2200	-	mcd	I_F=20mA
Radiant Intensity		2.5	4.0		mW/sr	
Forward Voltage	V_F	-	2.3	2.6	V	
		-	2.5	2.8	V	I_F=100mA 1/10 Duty@1KHz
Peak Wavelength	λ_P	-	525	-	nm	I_F=20mA
Spectral Bandwidth	$\Delta\lambda$	-	35	-	nm	
Reverse Current	I_R	-	-	10	μA	V_R=5V
View Angle	$2\theta_{1/2}$	-	120	-	deg	I_F=20mA

❶ Tolerance of Radiant Intensity/luminous Intensity is ±10%.

❷ Tolerance of Forward Voltage is ±0.03V.

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3. Product Binning

a). Luminous Intensity (Tolerance is ±10%)

BIN code	IR(mW/sr)		R(mcd)		G(mcd)		Condition
	Min.	Max.	Min.	Max.	Min.	Max.	
L09	1.5	3.5	260	360	1400	2000	I_F=20mA
L0A					2000	2800	

b). Peak Wavelength (Tolerance is ±1nm)

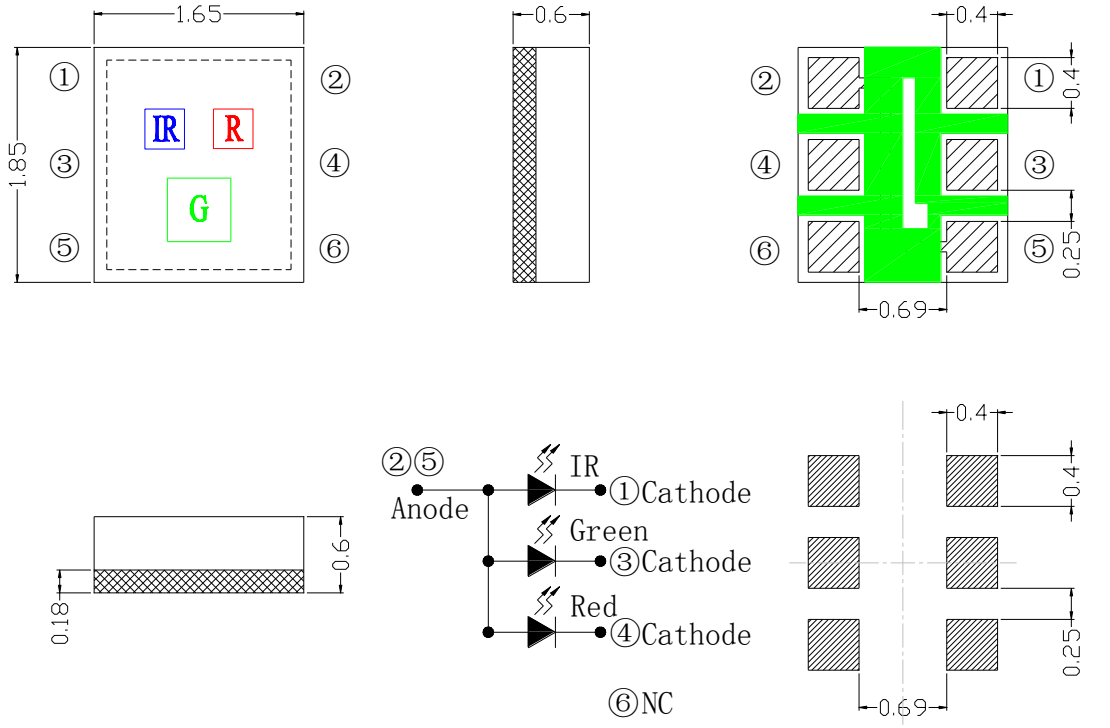
BIN code	IR(nm)		R(nm)		G(nm)		Condition
	Min.	Max.	Min.	Max.	Min.	Max.	
W01	930	950	650	670	515	520	I_F=20mA
W02					520	525	
W03					525	530	
W04					530	535	

c). Forward Voltage (Tolerance is ±0.05V)

BIN code	IR(V)		R(V)		G(V)		Condition
	Min.	Max.	Min.	Max.	Min.	Max.	
V02	1.1	1.5	1.8	2.3	2.2	2.4	I_F=20mA
V03					2.4	2.6	

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4. Package Outline Dimension



Suggestion Soldering Pad

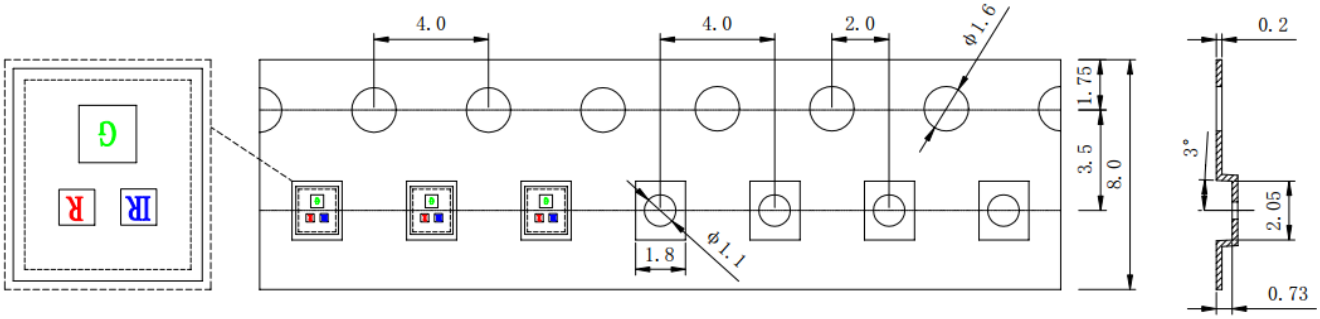
❶ All dimensions are in millimeters.

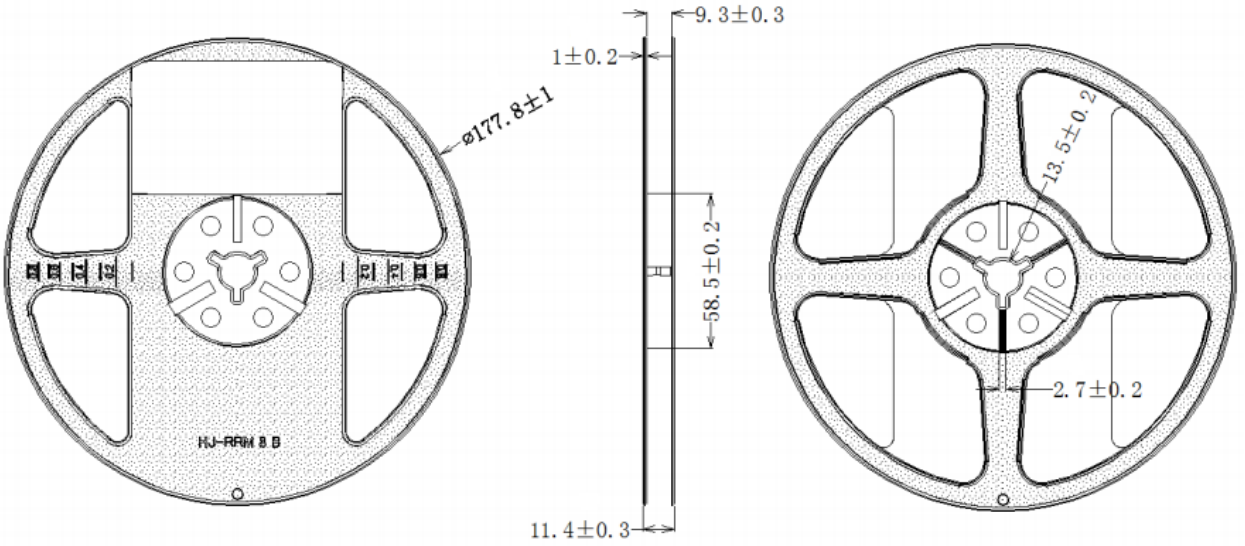
❷ Tolerance is ±0.1mm unless otherwise specified.

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5. Tapping specifications (unit: mm)

Loaded quantity: 4000 pcs/reel



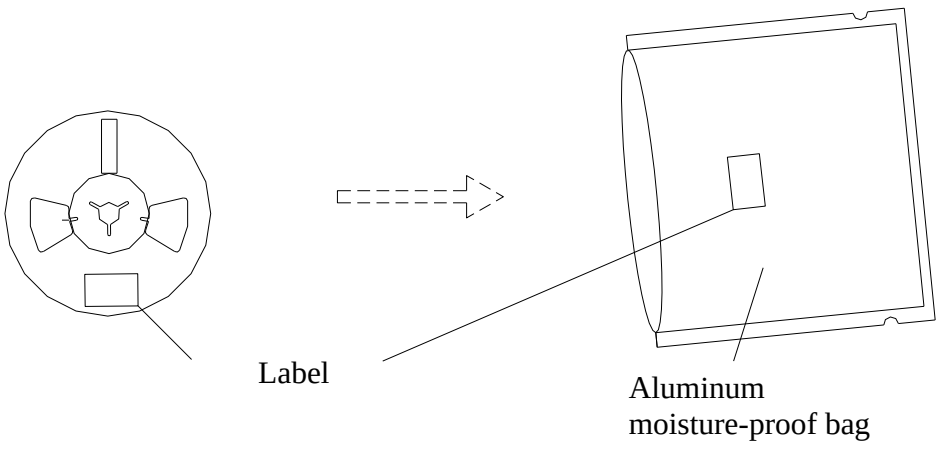


❶ All dimensions are in millimeters.

❷ Tolerance is ±0.1mm unless otherwise specified.

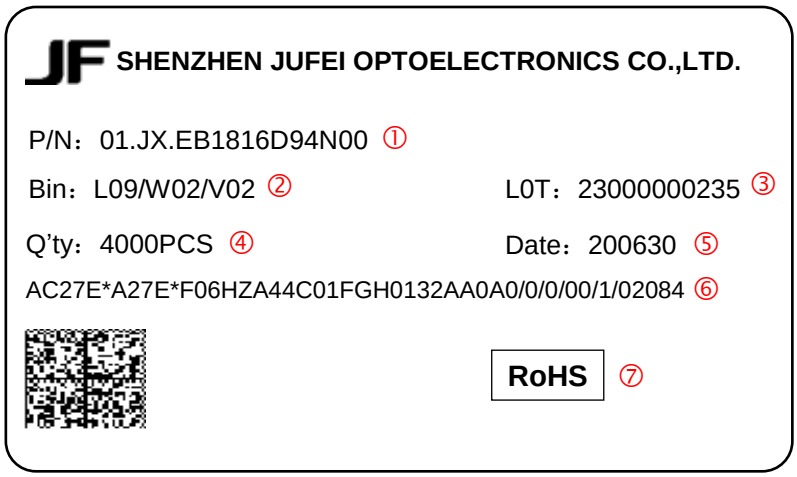
❸ Gewicht/Approx.weight:4.0±0.5mg.

6. Package Method:(unit: mm)



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7. Label description

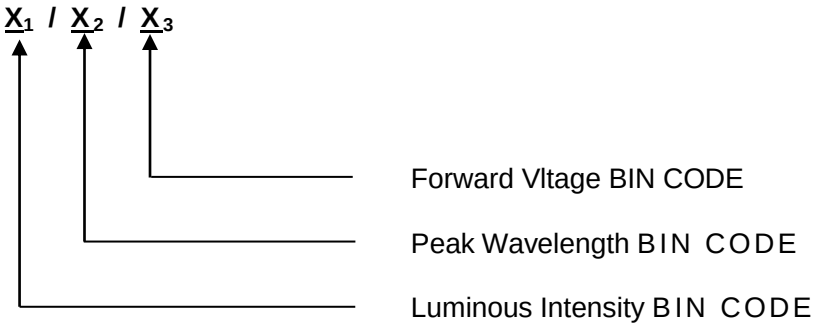


a) Part Number System description

$X_1X_2.X_3X_4.X_5X_6X_7X_8X_9X_{10}X_{11}X_{12}X_{13}X_{14}X_{15}$

Part Number Code	Description	Part Number	Value
X_1X_2	Production Type	01	Self-produce
X_3	Company	J	JF
X_4	Applications	X	Infrared
X_5	Product Type	E	Emitter
X_6	Power Range	B	0.1W~0.5W
X_7-X_{10}	Product Size	1816	1.8mm*1.6mm
X_{11}	Chip Type	D	IR chip
$X_{12}X_{13}$	Wavelength	94	940nm
X_{14}	Patent	N	--
X_{15}	Water code	00	--

b) BIN description

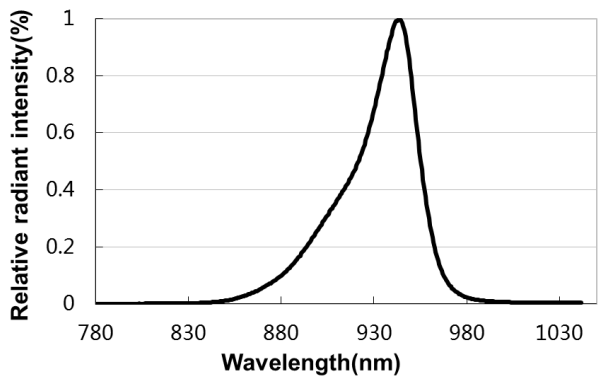


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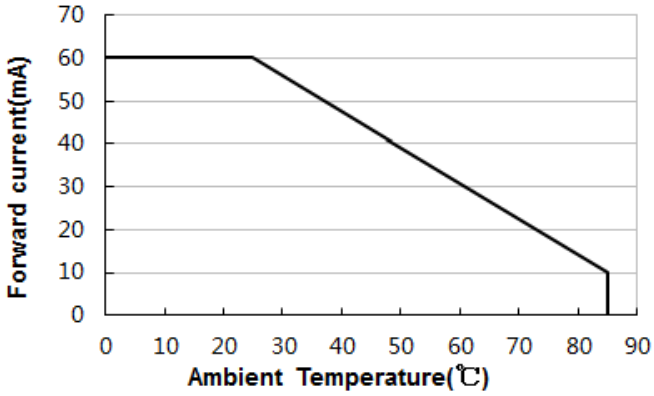
8. Typical Electro-Optical Characteristics Curves

Typical Electrical/Optical/Characteristics Curves for IR

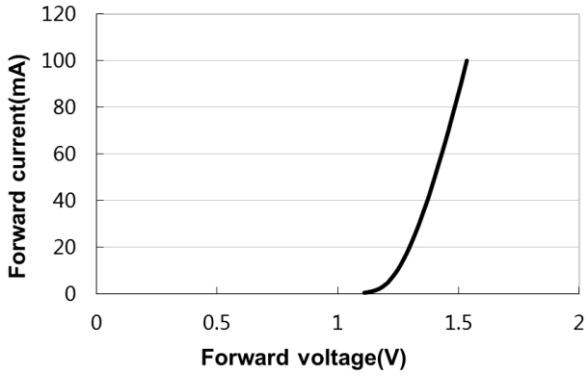
Spectrum



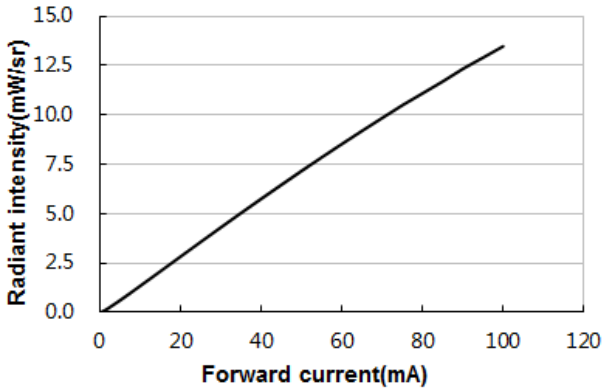
Forward current vs. Ambient temperature



Forward current vs.Forward voltage



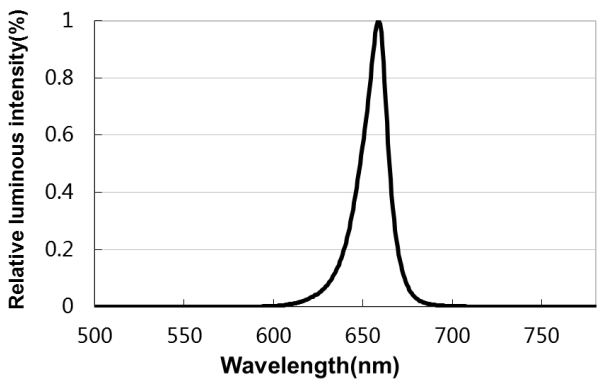
Radiant intensity vs.Forward current



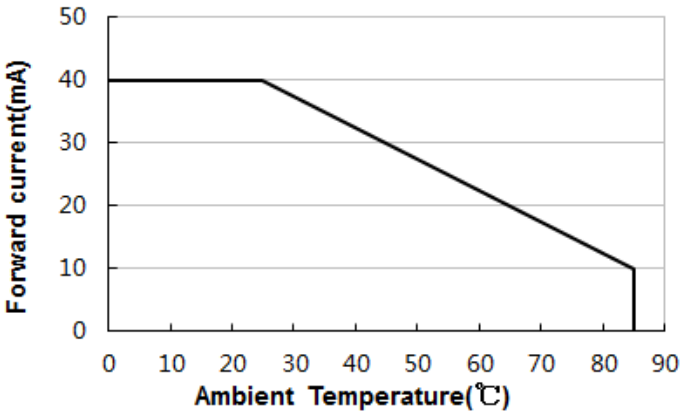
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Typical Electrical/Optical/Characteristics Curves for R

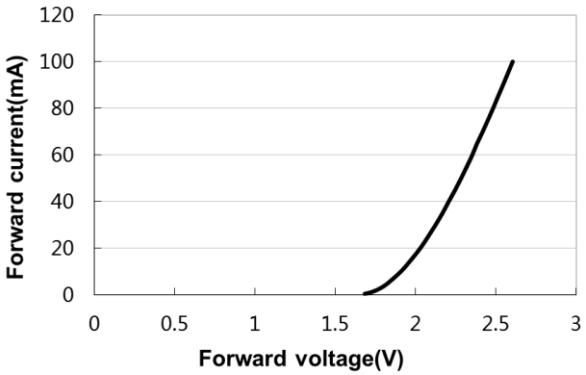
Spectrum



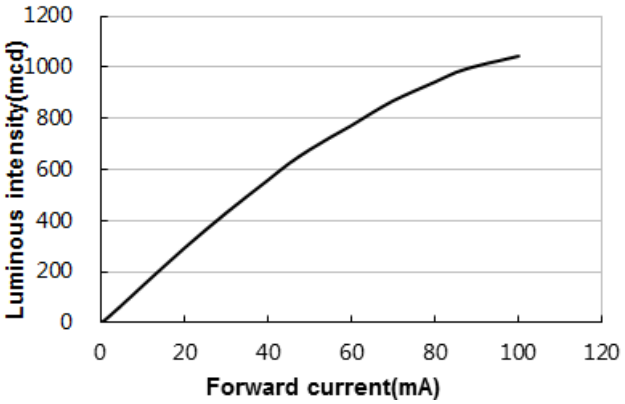
Forward current vs. Ambient temperature



Forward current vs.Forward voltage



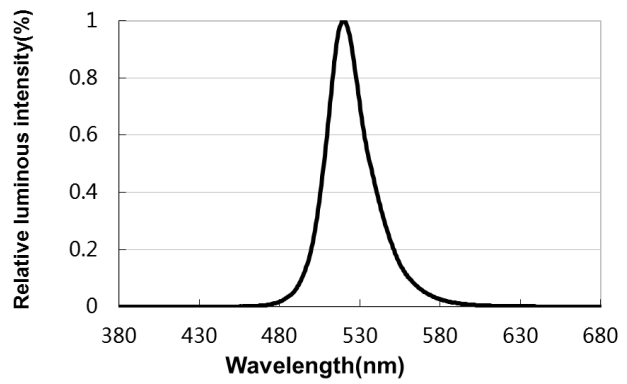
Luminous intensity vs.Forward current



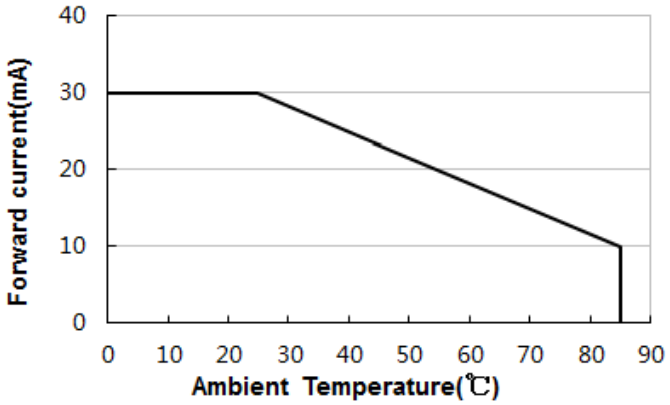
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Typical Electrical/Optical/Characteristics Curves for G

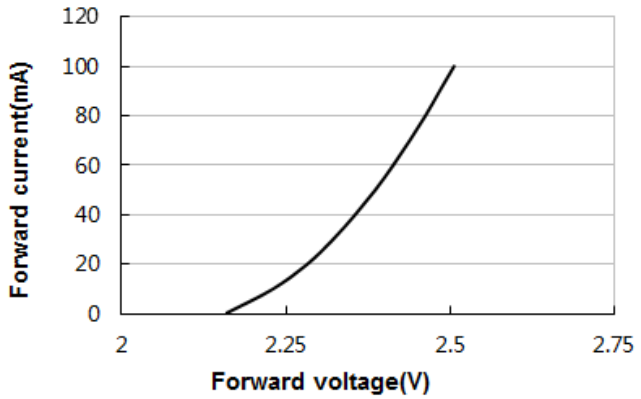
Spectrum



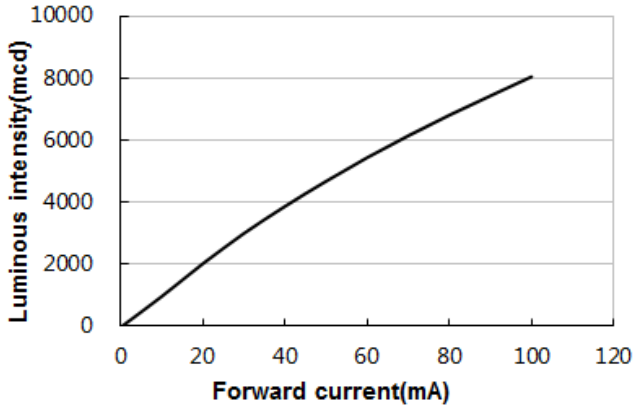
Forward current vs. Ambient temperature



Forward current vs.Forward voltage



Luminous intensity vs.Forward current



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9. Reliability test items and conditions

No.	Test Item	Applicable Standard	Test Conditions	Sample size	Ac/Re
1	Room Temperature Operating Life	JF Standard	If=DC20mA Temp: +25℃ Test time=1000hrs	22	0/1
2	Low Temperature Operating Life	JESD22-A108	If=DC20mA Temp: -40℃ Test time=1000hrs	22	0/1
3	High Temperature Operating Life	JESD22-A108	If=DC20mA Temp: +85℃ Test time=1000hrs	22	0/1
4	Wet High Temperature Operating Life	JESD22-A101 201	If=DC20mA Temp. =+60°C, RH=90% Test time=1000hrs	22	0/1
5	Pulse Operating Life	JESD22-A108	If=100mA Temp. =+85°C, 1/10 Duty@1KHz Test time=1000hrs	22	0/1
6	Thermal Shock	JEITA ED-4701 300 307	-40°C ~ +100°C 15min 5min 15min Test Time=100cycles	100	0/1
7	High Temperature Storage	JEITA-ED-4701 200 201	Temp: +100°C Test time=1000hrs	22	0/1
8	Low Temperature Storage	JEITA-ED-4701 200 202	Temp: -40°C Test time=1000hrs	22	0/1
9	Reflow Soldering	JEITA ED-4701 300 301	Operation heating: 260°C (Max.) within 10seconds. (Max.) 3reflows	22	0/1
10	ESD_{HBM}	MIL-STD-88E	R=1.5KΩ,C=100pF ±2000V	22	0/1

❶ Judgment criteria of failure for the reliability

❷ Intensity: Below 70% of initial values

❸ Vf: Over 20% of upper limit value

Note:

The tested LED have been returned to normal ambient conditions before testing

Measurement shall be taken within 2 hours

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10. Precautions for use

10.1 Soldering

SMD LED encapsulation is very flexible, outside force easily demolish radiant surface and plastic, As soldering , Please handle with care.

a. With No-clean Flux, according to reflow soldering cure condition when soldering, Reflow soldering should not be done more than two times, simultaneity you must insure clean on the radiant surface. Otherwise, foreign objects can affect radiant color.

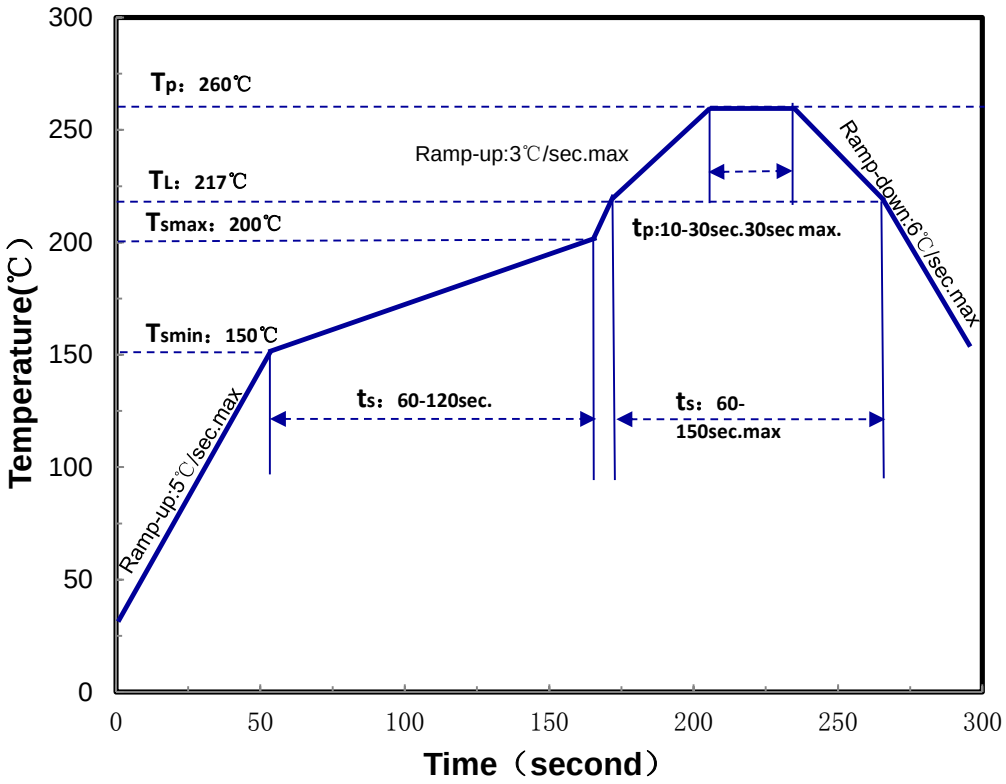
b. Don't process manual soldering except repair. Recommended to be soldered with 25W Anti-static iron, The temp. of the iron should be lower than 300℃ and soldering time should not be done more than three seconds, at the same time iron can't touch radiant surface and plastic.

c. Don't twist LED in course of manual soldering and experiment, Otherwise, the lights will not work possibly.

d. Please use the same BIN grade in one panel, and don't mix the difference BIN grade in one panel when soldering. Otherwise, it will cause a serious uneven color problem.

e. Please control the sulfur content of solder paste and PCB.

f. Pb-free solder temp.-time profile as below:260℃Max.



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Profile Feature	Lead Free Assembly
Temperature min（T_{smin}）	150℃
Temperature max(T_{smax})	200℃
Maximum time (t_s)from Tsmin to T_{smax}	120 seconds
Ramp-up(T_L to T_P)	3℃/sec
Liquids Temperature(T_L)	217℃
Maximum Time(t_L) Maintained T_L	150 seconds
Maximum Peak Package Body Temperature(T_p)	260℃
Time Within 5℃ of the Specified Temperature	10-30seconds
Maximum Ramp-Down Rate(T_p to T_L)	6℃/seconds
Maximum Time 25℃ to Peak Temperature	8minutes

10.2 Cleaning

a. Don’t be cleaned with ultrasonic. Recommended to be wiped with isopropyl alcohol or pure alcohol, wiping time should not be more than one minute. LED must be placed at room temperature for fifteen minutes before producing .you must insure clean on the radiant surface. Otherwise, foreign objects can affect radiant color.

b. LED can’t be in contact with acetate、trichloroethylene，acetone、sulfur、nitride、acid、alkali、salt. These matters can destroy LED.

10.3 Sealing

a. Sealing glue can’t contain sulfur，because these matter can affect fluorescence powder poisoning.

b. When using normal sealing glue, recommended will be operated life for 168hrs under normal temperature.

10.4 Storage

a. Don’t open the moisture proof bag before ready to use the LEDs.

b. The LEDs should be kept at 30℃ or less and 60%RH or less before opening the package. The max. period before opening the package is 1 year.

c. After opening the package, the LEDs should be kept at 30-35%RH or less, and it should be used within 3 days. If the LEDs should be kept at 30-35%RH or more, and it should be used within 4 hours.

d. If the LEDs be kept over the conditions of 20%, baking is required before mounting. Baking condition as below: 70±5℃ for 12 hours for bulk goods, 105±5℃ for 1 hours for roll goods.

e. The environment have no acid、alkali、corrosive gas、intensively shake and high magnetic field.

10.5 Static

a. Static and Peak surge voltage can destroy LED, Avoiding Instantaneous voltage when turn on or turn off the lights.

b. Please wear Anti-static wrist band、Anti-static glove、Anti-static shoes in the course of operation, and the equipment must be grounded.

c. After LED is be destroyed, leakage current increase obviously, and it will be forward voltage falling or failure lamp in the case of low current.

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10.6 Test

a. Customer must apply the current limiting resistor in the circuit so as to drive the LEDs within the rated current. Otherwise slight voltage shift maybe will cause big current change and burn out will happen.

b. Also, caution should be taken not to overload the LEDs with instantaneous high voltage at the turning ON and OFF of the circuit. Otherwise LED will be destroyed.

c. The reverse voltage mustn't exceed 5v when lighting on or testing the LED, otherwise, LEDs will be damaged.

10.7 Else

Radiant color of LEDs will be a little change with the current, recommended that LED is be used in series and resistance, when lighting, please don't see directly radiant surface of LED, otherwise LED will burn eye

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Revision History

Version: A0

Created by：

Version	Subjects(major change in previous version)	Date of change
RD	First public	11.30.2020
RD1	Change electrical-optical characteristics	01.27.2021
A0	Change product binning	06.12.2021

About Jufei@

The company production plant to the implementation of 10,000 purification, temperature and humidity, anti-static, the company introduced the most advanced SMD LED automatic production equipment, strict real integrated management system of ISO9001/TS16949, QC080000, ISO14001, OHSAS18001, and passed the CQC third-party certification; products by SGS, fully comply with the ROHS / REACH / halogen-free product environmental requirements. Jufei cost photoelectric LED products, widely recognized by the customers, has become the top ten brands of Chinese LED industry market customer satisfaction, and received the title of "Shenzhen Top Brand".

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